



CALICUT UNIVERSITY – FOUR-YEAR UNDER GRADUATE PROGRAMME (CU-FYUGP)

BSc PHYSICS HONOURS

Programme	B.Sc. Physics Honours				
Course Title	FUNDAMENTALS OF PHYSICS				
Type of Course	Core in Major				
Semester	I				
Academic Level	100 - 199				
Course Details	Credit	Lecture per week	Tutorial per week	Practical per week	Total Hours
	4	3	-	2	75
Pre-requisites	Fundamentals of vectors, calculus and kinematics.				
Course Summary	This course explores Newton's Laws of Motion and how they can be applied to solve different mechanical systems.				

Course Outcomes (CO):

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
CO1	Understand the concepts of Newton's Laws of Motion	U	C	Instructor-created exams / Quiz
CO2	Apply Newton's Laws of Motion to solve different mechanical systems	Ap	P	Instructor-created exams / Home Assignments
CO3	Apply work-energy theorem to solve different mechanical systems	Ap	P	Instructor-created exams / Home Assignments
CO4	Analyse conservative systems and solve them using the	An	P	Instructor-created exams / Home

	conservation of mechanical energy.			Assignments
CO5	Demonstrate critical thinking and problem-solving skills by applying the concepts and techniques learned to solve an extended set of real-world problems.	Ap	P	Seminar Presentation / Group Tutorial Work
CO6	Demonstrate skills to set up and perform experiments to test Newton's Laws of Motion and related concepts.	Ap	P	Practical Assignment / Observation of Practical Skills / Viva Voce
* - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C) # - Factual Knowledge(F), Conceptual Knowledge (C), Procedural Knowledge (P), Metacognitive Knowledge (M)				

Detailed Syllabus:

Module	Unit	Content	Hrs (45 +30)	Marks (70)
I	NEWTON'S LAWS OF MOTION		12	18
	1	Review of units, physical quantities and vectors	3	
	2	Force and Interactions	2	
	3	Newton's First Law	2	
	4	Newton's Second Law	2	
	5	Mass and Weight	1	
	6	Newton's Third Law	1	
	7	Free-Body Diagrams	1	
	Relevant topics of chapter 1 of Book 1; sections 4.1 – 4.6 of chapter 4 of Book 1 Self-Study: Chapters 1 – 3 of Book 1			

II	APPLYING NEWTON'S LAWS		14	20
	8	Using Newton's First Law: Particles in Equilibrium	3	
	9	Using Newton's Second Law: Dynamics of Particles	3	
	10	Apparent Weight and Apparent Weightlessness	1	
	11	Friction Forces	2	
	12	Fluid Resistance and Terminal Speed	1	
	13	Dynamics of Circular Motion	3	
	14	The Fundamental Forces of Nature	1	
	Sections 5.1 – 5.5 of chapter 5 of Book 1			
III	WORK AND KINETIC ENERGY		8	14
	15	Work	1	
	16	Kinetic Energy and the Work – Energy Theorem	3	
	17	Work and Energy with Varying Forces	3	
	18	Power	1	
	Sections 6.1 – 6.4 of chapter 6 of Book 1			
IV	POTENTIAL ENERGY AND ENERGY CONSERVATION		11	18
	19	Gravitational Potential Energy	3	
	20	Elastic Potential Energy	2	
	21	Conservative and Nonconservative Forces	2	
	22	Force and Potential Energy	2	
	23	Energy Diagrams	2	

	Sections 7.1 – 7.5 of chapter 7 of Book 1									
V	PRACTICALS	30								
	Conduct any 5 experiments from the given list and 1 additional experiment, decided by the teacher-in-charge, related to the content of the course. The 6 th experiment may also be selected from the given list. Other experiments listed here may be used as demonstrations of the concepts taught in the course.									
	• Error Analysis: Lecture/ Tutorial/ Seminar: 2 hrs. • Theory of experiments 1 and 2 can be given as Assignment/ Seminar. • Plot the graphs using GeoGebra. FitLine function may be used to get the slope. • Smartphones are exclusively intended for educational lab use. Necessary care should be taken to safeguard them during the experiments. • Smartphone experiments primarily serve demonstration purposes, with result accuracy contingent upon the precision of phone sensors and experimental setups.									
	<table border="1"> <tr> <td>1</td><td> Young's Modulus of the Material of a Given Bar: Uniform Bending • Use an optic lever and telescope. Take measurements for a minimum of two lengths. Obtain the elevation (e) from the shift (s) in the telescope reading and calculate Y from it. • For each length of the bar, plot the load-elevation graph (using GeoGebra) and obtain m/e, and then calculate Y from it. • Estimate the random error in the measurements and the error of the result using propagation of the error formulae. </td><td></td><td></td></tr> <tr> <td>2</td><td> Young's Modulus of the Material of a Given Bar: Non-Uniform Bending • Use a pin and a microscope. Take measurements for a minimum of two lengths. Obtain the depression (e) from the shift in the microscope reading and calculate Y from it. </td><td></td><td></td></tr> </table>	1	Young's Modulus of the Material of a Given Bar: Uniform Bending • Use an optic lever and telescope. Take measurements for a minimum of two lengths. Obtain the elevation (e) from the shift (s) in the telescope reading and calculate Y from it. • For each length of the bar, plot the load-elevation graph (using GeoGebra) and obtain m/e, and then calculate Y from it. • Estimate the random error in the measurements and the error of the result using propagation of the error formulae.			2	Young's Modulus of the Material of a Given Bar: Non-Uniform Bending • Use a pin and a microscope. Take measurements for a minimum of two lengths. Obtain the depression (e) from the shift in the microscope reading and calculate Y from it.			
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	3	Verification of Newton's First Law: Equilibrium of a Particle - Analyze the two dimensional equilibrium problems using spring / digital force gauges. - Hang a weight from a chain that is linked at the ring to two other chains, one fastened to the ceiling and the other to the wall. Example 5.3 of Book 1. - Measure the angle between the chain from the ceiling and the horizontal and the tension in each of the three chains using spring/digital force gauges and verify with the theoretical predictions. https://www.youtube.com/watch?v=XI7E32BROp0		
	4	Coefficient of Static Friction. - Determine the coefficient of static friction between a wooden block and a wooden plane. - Measure the angle at which the wooden block just starts to slide down an inclined wooden plane and hence calculate the static friction coefficient. https://www.youtube.com/watch?v=gt8mr6pFSFE OR - Place the wooden block on a wooden plane surface and add mass to the pan attached to the block using a string through a frictionless pulley. - Find the mass required to initiate the sliding of the block. - Different trials can be done by adding mass on the top of the block and hence determine the coefficient of static friction. - Example 5.13 of Book 1. https://www.youtube.com/watch?v=MSV6VafiUF4&t=443s		
	5	Acceleration of a Freely Falling Body - Use the smartphone acoustic stopwatch to determine the duration of a free fall. - Measure the time of flight of a steel ball for different heights and plot a graph of distance versus. time squared (s vs. t^2). Determine g from the graph. - Experiment 2 of Book 2. - Phyphox app may be used. https://phyphox.org/experiment/free-fall-2/		

		OR Use ExpEyes kit, electromagnet, and contact sensor to determine the duration of a free fall. https://expeyes.in/experiments/mechanics/tof.html		
	6	Verification of the Relation of Angular Velocity and Centrifugal Acceleration - Use the smartphone gyroscope and the accelerometer. - Attach the smartphone to some rotating arrangements and record the data from the gyroscope and accelerometer. - Plot angular velocity versus acceleration and verify the relation. - Experiment 18 of Book 2. - Phyphox app may be used. https://phyphox.org/experiment/centrifugal-acceleration/		
	7	Analysis of Bouncing Balls to Determine Gravitational Acceleration and Coefficient of Restitution. - After doing the experiment, the student should be able to understand the concept of inelastic collision. - Measure the time interval between successive bounces using a digital acoustic stopwatch and hence calculate g and coefficient of restitution - Experiment 12 of Book 2 and section 3.3 of Book 1 - Phyphox app may be used. https://phyphox.org/experiment/inelastic-collision/		
	8	The Nearly Parabolic Trajectories of a Bouncing Ball - Perform Experiment 7 using Tracker tool. - Track the ball and plot the time versus position graph. - Measure the time interval between successive bounces and hence calculate g and coefficient of restitution. - Experiment 12 of Book 2 and section 3.3 of Book 1 - Tracker Autotracker Tutorial: https://www.youtube.com/watch?v=Dn0Zz7rtkZw		
	9	Verification of Newton's Second Law: Atwood's Machine - Determine the relationship between the vertical acceleration and the mass difference, using a smartphone accelerometer. - The vertical acceleration is registered using the built-in accelerometer of the smartphone. - By redistributing the masses of the supports, a linear relationship between the mass difference and the vertical acceleration is obtained.		

		- Experiment 8 of Book 2. https://phyphox.org/experiment/acceleration-without-g/		
	10	Analysis of Air Resistance and Terminal Speed to Determine the Drag Coefficient. - Record the motion of a light weight paper cup and analyse it with Tracker tool (https://physlets.org/tracker/). - Plot acceleration, velocity, and position with time. - Repeat the experiment with different mass (by simply stacking the paper cups) - Determine the Drag Coefficient - Experiment 27 of Book 2. https://www.youtube.com/watch?v=iujzK3uH1Yc		
	11	Projectile Motion: Kinematics - Analyse projectile motion as a combination of horizontal motion with constant velocity and vertical motion with constant acceleration. - Drop two balls from a height, one from rest, and other simultaneously projected horizontally. - Analyse the motion of both in the Tracker tool. - Section 3.3 of Book 1 https://www.youtube.com/watch?v=zMF4CD7i3hg https://www.youtube.com/watch?v=Ml01anodoDE https://www.youtube.com/watch?v=5I0NLNthJGc		
	12	Projectile Motion: Energy Conservation - Analyse the motion of the tossing ball / projectile in the Tracker tool. - Plot time versus the x-and y-components of velocity and acceleration. - Also plot the kinetic energy, potential energy (build data using define tool) and total energy. https://www.youtube.com/watch?v=x0AWRLvgB28 https://www.youtube.com/watch?v=i07HeUWo8xc		

Books and References:

- University Physics with Modern Physics (Edn.15) by Young & Freedman (Book 1)
- Smartphones as Mobile Minilabs in Physics(Edn. 1) by Jochen Kuhn & Patrik Vogt, Springer, (Book 2)
- <https://phyphox.org/>
- <https://physlets.org/tracker/>
- B.Sc Practical Physics by C L Arora
- Practical Physics by S L Gupta & V Kumar

7. Fundamentals of Physics by David Halliday, Robert Resnick and Jearl Walker
8. Physics for Scientists and Engineers by Paul A. Tipler and Gene Mosca
9. Fundamentals of Physics by J. Richard Christman and William J. Francis
10. NPTEL video lectures: <https://nptel.ac.in/courses/115106090>

Mapping of COs with PSOs and POs :

	PSO 1	PSO 2	PSO 3	PSO 4	PSO 5	PSO 6	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO 1	3	0	2	0	2	1	3	3	1	1	2	2	1
CO 2	3	2	2	0	1	2	3	3	3	1	3	3	3
CO 3	3	2	2	0	1	2	3	3	3	1	3	3	3
CO 4	3	2	2	3	1	2	3	3	3	1	3	3	3
CO 5	3	2	3	0	3	3	3	3	3	1	3	3	3
CO 6	3	3	3	3	1	3	3	3	3	1	3	3	3

Correlation Levels:

Level	Correlation
0	Nil
1	Slightly / Low
2	Moderate / Medium
3	Substantial / High

Assessment Rubrics:

- Quiz / Discussion / Seminar
- Internal Theory / Practical Exam
- Assignments / Viva
- End Semester Exam (70%)

Mapping of COs to Assessment Rubrics

	Internal Theory / Practical Exam	Assignment / Viva	Practical Skill Evaluation	End Semester Examinations
CO 1	✓	✓		✓
CO 2	✓	✓		✓
CO 3	✓	✓		✓
CO 4	✓	✓		✓
CO 5	✓	✓		✓
CO 6		✓	✓	